

KALAJOKI ULKOKALLA, Finland

WMO: 029070

Lat: 64.331N

Lon: 23.446E

Elev: 5

StdP: 101.26

Time Zone: 2.00 (EUE)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-19.7	-16.1	-21.8	0.5	-19.9	-17.9	0.8	-16.1	17.6	-1.0	16.3	-0.2	3.9	150	0.870

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	2.9	21.5	19.2	20.0	17.7	18.7	16.8	19.4	21.3	18.0	19.6	17.0	18.5	4.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.5	13.4	20.9	17.3	12.4	19.3	16.2	11.5	18.1	55.4	21.5	51.0	19.7	47.8	18.6	22.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.0	14.5	13.2	DB	-19.3	22.6	5.3	2.4	-23.1	24.3	-26.2	25.7	-29.2	27.0	-33.1	28.7	(4)
(5)			WB	-19.5	19.5	5.2	1.7	-23.3	20.7	-26.4	21.7	-29.3	22.6	-33.1	23.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.4	-4.4	-5.6	-3.7	0.5	5.0	10.4	15.5	15.3	11.6	6.2	2.3	-0.5	(6)	
	DBStd	7.99	5.14	5.82	4.34	2.11	2.58	2.54	2.86	2.51	2.32	2.68	3.13	4.03	(7)	
	HDD10.0	2462	446	438	424	285	157	26	1	1	11	119	231	326	(8)	
	HDD18.3	5084	704	671	682	535	414	239	98	98	201	376	481	584	(9)	
	CDD10.0	433	0	0	0	0	1	36	171	164	59	1	0	0	(10)	
	CDD18.3	13	0	0	0	0	0	0	9	4	0	0	0	0	(11)	
	CDH23.3	10	0	0	0	0	0	0	6	4	0	0	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	6.9	7.9	7.2	6.3	5.5	5.8	5.7	5.2	6.2	7.1	8.0	8.6	8.8	(14)	
(15) Precipitation	PrecAvg	742	66	53	49	41	49	63	79	75	61	67	68	73	(15)	
	PrecMax	955	130	101	81	90	93	112	128	122	119	124	137	118	(16)	
	PrecMin	570	18	6	9	13	12	22	20	28	26	16	8	27	(17)	
	PrecStd	95	29	26	18	20	23	24	30	28	30	28	28	26	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.4	3.1	3.5	6.8	13.2	17.7	24.2	23.5	17.3	12.2	8.8	5.3	(19)	
		MCWB	2.4	1.9	1.9	4.8	11.5	15.1	20.6	20.6	15.1	10.8	7.6	4.3	(20)	
	2%	DB	2.5	2.1	2.5	4.8	11.0	16.2	22.4	21.0	16.2	11.3	7.9	4.7	(21)	
		MCWB	1.8	1.1	1.3	3.1	9.3	14.1	20.0	18.3	14.4	10.2	7.1	3.9	(22)	
	5%	DB	1.9	1.2	1.9	3.7	9.5	15.0	20.8	19.6	15.4	10.5	7.1	4.2	(23)	
		MCWB	1.3	0.6	0.9	2.3	8.1	13.4	18.6	17.4	13.9	9.5	6.3	3.4	(24)	
	10%	DB	1.3	0.7	1.3	2.9	8.2	13.8	19.5	18.6	14.7	9.7	6.2	3.6	(25)	
		MCWB	0.8	0.1	0.5	1.9	6.9	12.4	17.5	16.6	13.3	8.6	5.4	2.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.9	2.1	2.2	5.2	12.3	15.7	21.3	20.8	15.7	11.3	8.0	4.7	(27)	
		MCDB	3.1	2.8	2.9	6.1	12.5	16.8	23.2	23.2	16.6	11.8	8.5	4.9	(28)	
	2%	WB	2.0	1.3	1.6	3.2	10.0	14.5	20.2	18.7	14.8	10.4	7.3	4.2	(29)	
		MCDB	2.3	1.7	2.2	4.3	10.8	15.7	22.1	20.4	15.8	11.1	7.8	4.6	(30)	
	5%	WB	1.4	0.7	1.1	2.6	8.3	13.7	18.7	17.7	14.2	9.6	6.5	3.5	(31)	
		MCDB	1.8	1.1	1.6	3.4	9.3	14.8	20.5	19.2	15.1	10.3	7.0	4.1	(32)	
	10%	WB	0.8	0.2	0.6	2.0	7.0	12.5	17.7	16.9	13.5	8.8	5.6	2.9	(33)	
		MCDB	1.3	0.7	1.2	2.8	8.1	13.6	19.2	18.3	14.6	9.5	6.1	3.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	3.5	3.7	4.1	3.1	3.1	3.0	2.9	2.7	2.4	2.3	2.4	2.6	(35)	
		MCDBR	2.5	2.9	3.0	3.8	5.3	4.2	4.4	3.8	3.0	2.4	2.4	2.5	(36)	
	5% WB	MCWBR	2.5	2.7	2.6	2.8	3.9	3.1	3.0	2.7	2.5	2.4	2.4	2.4	(37)	
		MCWBR	2.6	2.8	2.8	3.4	5.2	3.9	4.0	3.4	2.7	2.4	2.4	2.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.220	0.282	0.308	0.346	0.348	0.351	0.370	0.356	0.332	0.311	0.348	0.350	(40)	
		taud	1.995	2.128	2.171	2.217	2.437	2.488	2.457	2.486	2.531	2.369	2.559	2.569	(41)	
	Ebn at Noon		330	612	766	822	864	868	836	815	762	593	303	147	(42)	
		Edh at Noon	26	61	90	109	97	94	95	84	66	50	23	14	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.12	0.68	2.23	3.94	5.33	5.96	5.38	4.05	2.25	0.81	0.17	0.04	(44)	
	RadStd		0.01	0.14	0.26	0.35	0.48	0.49	0.48	0.34	0.23	0.15	0.03	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (44 neighbors)	+0.42	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page